AzurESky	AZURESKY PTY LTD	Document # AZCA-007-2	Revision# 0
Subject	NOMINATED REPRESANTATIVE TECHNICAL ASSESSMENT		

NOMINATED REPRESENTATIVE TRAINING AND ASSESSMENT

Name and Surname: NonMamhla Mshbl

Date: 13/11/2025

Total = 52 marks Questions = 32

SECTION A: Roles, Responsibilities and SANAS Requirements (R03-10 & R04) (12 marks)

1. Explain the primary responsibilities of a Technical Signatory as outlined in R03-10. How does this role ensure the technical validity of test results? (2)

Answer:
It is to report results issued under the CABs accreditation & whose signature confirms validity of content on CABs certificates or reports. 2
They confirm technical validity of test reports by confirming that validated methods trained personnel verified & calibrated equipment; they meet the requirement of ISO 17025 & assure integrity of data.

2. Describe the process your laboratory follows to nominate and approve a new Technical Signatory. Which documentation and evidence must be submitted to SANAS? (2)

Answer:
Competency evaluation, witnessing of competence are conducted & management endorsement. Evidence such as qualification certificates, CV's, training records & an updated organogram are submitted to SANAS.

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3. Discuss three qualification or experience requirements that must be met before approval as a Technical Signatory under R03-10. (2)

Answer:
* A Tertiary qualification or equivalent technical experience in the test fields.
* Demonstrated competence
* Experience (Technical experience) in performing interpreting & authorising results should be > 3 years.

4. Explain the difference between a Nominated Representative and a Technical Signatory according to R03-10, and why these roles must be clearly defined. (2)

Answer:
An NR is important/primary role to contact between the CAB & SANAS
A TS technical competence as confirms by the CAB & whose signature confers validity of the content on reports & certificates.

5. Describe how the Technical Signatory ensures correct use of the SANAS logo and reference to accreditation in reports, per R04. Give one example of incorrect usage. (2)

Answer:
TS ensures that the SANAS logo & accreditation reference appear only on reports for accredited methods with correct wording such as "SANAS Accredited Testing laboratory No.."
An example of incorrect usage include continuing to use the logo after the accreditation has been suspended, expired or withdrawn.

6. What other document(s) other than the 7 primary SANAS document must the NR be conversant with? (1)

Reports must clearly reference the accreditation by including the SANAS logo, accreditation number & method identifiers that align with the accreditation scope.

7. Outline the procedure for referencing SANAS accreditation on laboratory reports to maintain traceability. (2)

Answer:
Cross-referencing of results to method codes & revision numbers ensure full traceability & compliance. 2

SECTION B: Suspension, Withdrawal and Reinstatement (R51-10) (8 marks)

8. Under what circumstances may SANAS suspend or reduce a laboratory's scope? Provide an example. (2)

Answer:
2
In a case where a CAB fails to meet accreditation requirements eg. failing to close / address nonconformities recorded after the assessment.

9. State the laboratory's obligations during suspension or withdrawal, covering both administrative and technical aspects. (2)

Answer:
2
The CAB must cease to use SANAS logo; halt released of accredited results, inform affected clients & maintain records.
Testing of the CAB must continue & results must be clearly marked as non-accredited.

answer: _____

10. Describe the process and evidence required for reinstatement following suspension. (2)

Answer:
The Laboratory must submit corrective evidence addressing the root cause for the nonconformities & undergo a SANAS reassessment or document review confirming full compliance.

11. What period is set out in SANAS R 03 that if a NR can be absent, the CAB can be required to assign an interim NR?(1)

Answer:
A period longer than six weeks but not exceeding 6 (six) months.

12. Discuss the possible impacts of accreditation withdrawal on laboratory credibility and client confidence. (2)

Answer:
- It can lead to legal issues where legal agreement may be suspended until SANAS accreditation status is reinstated.
- Loss of business, Clients may stop using the labs services if they can no longer rely on the results.

SECTION C: Proficiency Testing and Interlaboratory Comparison (R80-06) (8 marks)

13. Explain the importance of proficiency testing (PT) participation for accredited laboratories and how it supports confidence in results. (2)

Answer:
PT participation demonstrates that the lab can produce accurate, comparable results with other labs.
It supports confidence of both client & accreditation body.

2

14. List criteria to consider when selecting a suitable PT or interlaboratory comparison provider. (2)

Answer:
PT provider's accreditation status
Appropriate frequency & statistical evaluation

15. How should a Technical Signatory respond to unsatisfactory PT performance? Describe the steps for investigation and correction. (2)

Answer:
* Determine/analyse the root cause
* Implement corrective action
* Perform retest/method checks
* Document the outcome in PT review records.

16. Discuss how PT data can be used for ongoing staff competence verification. (2)

Answer:
PT results form part of competence evaluation by confirming staff's analytical proficiency. Consistent satisfactory performance validates ongoing technical competence & may reduce the need for additional witness testing.

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SECTION D: Calibration and Intermediate Checks (TR25) (10 marks)

17. Define "intermediate check" per TR25 and explain its purpose in maintaining measurement confidence. (2)

Answer:
An intermediate check is a performance check carried out between calibration to verify ongoing accuracy & stability of equipment.
It maintains confidence in measuring results.

18. Give three examples of coal-testing instruments requiring intermediate checks and describe what parameter each check verifies. (2)

Answer:
- Analytical balance: checks with certified weights, verifies accuracy + linearity
- Oven: check with calibrated thermometer, verifies temperature control.
- Bomb calorimeter: checks with benzoic acid; verify energy calibration + repeatability

19. Explain the difference between calibration and verification, giving one laboratory example. (2)

Answer:
- Calibration establishes a relationship between instrument readings + reference standards.
- Verification confirms that the instrument still meets performance requirements without altering calibration.

eg. Oven, verification of temperature control, this uses calibrated thermometer.

20. Describe how the Technical Signatory ensures calibration certificates meet traceability requirements in TR25. (2)

Answer:
A TS ensures that calibration meets traceability requirements by checking calibration certificates that they include: uncertainty statements, environmental conditions, correction factors, traceability to national/international standards.

21. Discuss the impact of using equipment that has missed scheduled checks on test validity and accreditation compliance. (2)

Answer:
* Results may be rejected leading to reputation loss or loss of customer trust
* It may lead to financial loss in the sense that final lab may be suspended due to unsatisfactory results. 2

SECTION E: Method Validation and Verification (TR26) (8 marks)

22. Differentiate between validation and verification of analytical methods as per TR26 and indicate when each applies. (2)

Answer:
- Validation: applies to new/modified methods to confirm suitability for intended use
- Verification: applies to standard methods to confirm that they perform as expected in the laboratory. 2

23. Identify three performance characteristics evaluated during method validation under TR26. (2)

Accuracy/precision
Limit of detection/quantification
Robustness & uncertainty 2

Answer:

24. Explain how the Technical Signatory ensures validated methods remain fit for purpose, citing two review mechanisms. (2)

Answer:
- Periodic method review & trending of QC results
- Ongoing performance verification (control charts, PTs, RMs) Any drift triggers re-evaluation or calibration.

25. Describe the link between uncertainty of measurement and method validation, and its importance in coal testing decisions. (2)

Answer:
Uncertainty of quantifies the range of confidence around the result. In coal testing it determines the comparability of results between labs & influences compliance decisions. It forms part of method validation evidence.

SECTION F: Technical Competence and Laboratory Operations (4 marks)

26. Explain how the Technical Signatory ensures accuracy and repeatability in: Ash Determination, Calorific Value, Inherent Moisture, ISO Volatile Matter, and Sample Preparation. (2)

Answer:
- Controlled sample preparation (consistent particle size + Moisture)

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- Use of calibrated equipment + verified methods
- Replicate testing, QC check, & analyst competency evaluation for Ash, CV, IM & VM analyses

27. Reflect on how SANAS documents R03-10 to TR26 collectively promote data integrity and continual improvement in an accredited coal laboratory. (2)

Answer:
R03-10 defines competence roles; R04 controls reference to accreditation; TR25 maintains calibration integrity; TR26 ensures validated methods. Together they embed data integrity, traceability & continual improvement across the laboratory.

SECTION G: Applied Analytical Calculations (Laboratory Data Exercise) (10 marks total – 5 questions × 2 marks)

28. Convert the Volatile Matter result from air-dry to dry-basis. (2)

Answer:
eg. 20,43 % VM (AD), IM for 3,20 %

$$\therefore \frac{20,43 \times 100}{(100 - 3,2)}$$

$$= 21,10\ \%\ VM.$$

29. Calculate the Total Moisture %. (2)

Answer: Step 1: Determine surface moisture (As received)
Step 2: " Residual moisture (Air dried)
Step 3: Convert RM from AD to as received.
Step 4: Total Moisture % = Surface Moisture (As received) + Residual Moisture (As received)

30. Convert the % Ash result from air-dry to as-received basis. (2)

eg. 30,41 % AD ; TM = 7,14 %

$$Ash\ (AR) = \frac{Ash\ (AD) \times (100 - TM)}{100}$$

$$= \frac{30,41 \times (100 - 7,14)}{100}$$

$$= 28,23\ \%$$

Answer:

31. Convert the Calorific Value from MJ/kg to kcal/kg. (2)

Answer: 1 MJ/kg = 239,0057 kcal/kg

∴ ~~20,143 MJ/kg~~ $_{N.P.}$ X 239,0057 25.52 X 239.0057

= 4882,88 kcal/kg. 1

32. Apply the appropriate correction for Volatile Matter and explain its purpose. (2)

Answer:

VM results must ~~be~~ corrected for Inherent Moisture to obtain values on a dry basis, ensuring comparability & accuracy in proximate analysis. The correction eliminates influence of Moisture content on volatile yield & calorific evaluations.

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ASSESSOR'S MARKING SHEET

Use this section to record marks and comments for each question.

Question No.	Mark Awarded	Assessor Comments
1	2	
2	2	
3	2	
4	2	
5	2	
6	1	
7	2	
8	2	
9	2	
10	2	
11	1	
12	2	
13	2	
14	2	
15	2	
16	2	
17	2	
18	2	
19	2	
20	2	
21	2	
22	2	
23	2	
24	2	
25	2	
26	2	
27	2	
28	2	
29		
30	2	
31	1	Incorrect value for ρ_V used but conversion is correct.
32	2	

51/52

98 6/10